Seasonal vector density and disease incidence patterns of malaria in an area of Zimbabwe

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Abstract

Anopheline mosquitoes were studied in an area of Zimbabwe to correlate their population densities with seasonality in malaria incidence. Four trapping methods were employed for adult mosquitoes: pit traps, outdoor and indoor man-baited nets and a sheep-baited net. *Anopheles funestus* exhibited two population peaks during the year, one in March and the other in September/October. The following over-all proportions of the *An. gambiae* complex were recorded: *An. gambiae* Giles (2·5%), *An. arabiensis* Patton (14·3%) and *An. quadriannulatus* Theobald (83·2%). Peak populations were observed in March, with most *An. gambiae* and *An. arabiensis* preferring outdoor man-baited nets. Approximately equal preference for outdoor nets and pit traps was observed for *An. quadriannulatus*, which was the only species present throughout the year. The implications of these findings for malaria transmission, which showed a peak in April, and the suitability of the sampling methods employed are discussed.

Introduction

Malaria remains an important disease of the tropics, affecting most Third World Countries, where little progress has been made in combatting it. In Europe the disease has been eradicated although occasionally imported cases are reported (LEGER *et al.*, 1980; BRUCE-CHWATT & ZULUETA, 1980).

In Zimbabwe, early studies by LEESON (1931) and later by Hadjinicolau (1961a, b, unpublished report) showed *An. gambiae* (sensu lato) to be an important vector of malaria. The *An. gambiae* species complex was unknown at the time and therefore the vectorial contributions of each of the sibling species were not differentiated, although MAHON *et al.* (1976) reported the four sibling species *An. arabiensis*, *An. gambiae*, *An. merus* and *An. quadriannulatus* to be present in Zimbabwe. The first three of these species but not the fourth, are known vectors of malaria in other parts of Africa (WHITE, 1974). *Plasmodium falciparum* is the most common malaria parasite in Zimbabwe, accounting for roughly 97% of diagnosed cases. Malaria endemicity in Zimbabwe varies from holo- or hyperendemic in the northern and southern lowveld to hypoendemic in the central high altitude region. Malaria transmission is not considered to occur regularly in areas above the 1200m contour. The latter, as with much of Zimbabwe below this altitude, is an unstable malaria zone where sporadic epidemics occur from time to time, probably as a result of vectors moving from the lower altitude zones into these upland areas whenever weather conditions encourage their breeding and increased longevity.

Very little information has been published concerning malaria transmission in Zimbabwe. TAYLOR & MUTAMBU (1981) have shown that the disease is markedly seasonal but no attempt has been made to correlate this seasonality with vector populations.

Against this background studies were undertaken at Chitengu village (820 metres altitude) in the Nyaguwe/Mazowe Valleys, an area intermediate between high and low altitude zones of Zimbabwe that was believed to harbour *An. gambiae s.l.* breeding all year round. The objectives of the study were to monitor the seasonal population densities and age-structure of malaria vectors in order to correlate these findings with the local malaria incidence, transmission patterns and climatic factors.

Study Site and Methods

Fig. 1 shows a sketch map of the village of Chitengu, situated some 80km north-east of Harare (Zimbabwe) at 17° 13'S × 31°45'E, and at an altitude of 820m above sea level. The village complex comprises about 30 huts scattered over an area of nearly one km^2. The village population is about 120 people. All the cattle (about 40) are kept in one pen overnight. The Mazowe and Nyaguwe Rivers have their confluence less than 1·0km west of the village.

Inhabitants of the village engage in gold searching on the river bed during the dry season and this leaves behind puddles of clear water. These puddles provide ideal breeding sites for anopheline mosquitoes before the onset of the early rains in November. The presence of two large rivers in the vicinity of the site always provided plenty of larval breeding sites when the water levels were low.

During the study, the area received sporadic rainfall from late December until April. Rainfall from the Shamva Experimental Station (Fig. 2a) is representative of the study area. After the rivers were flooded, larval breeding shifted away from the river bed to terrestrial permanent and semi-permanent pools and water puddles. Throughout the duration of the study, the wind direction was predominantly north-westerly.

Sampling procedure

Sampling of the *An. gambiae* complex was carried out over 14 days of each month, from November 1980 until October 1981. Four trapping methods were used. A series of 13 pit traps (later reduced to 12 due to flooding of one) of the MUIRHEAD-THOMSON (1958) type were set up on the western periphery of the village bordering the rivers (see Fig. 1). These were sampled each morning (9-10 a.m.) for resting mosquitoes. Two outdoor man-baited nets were set up, one at the base camp and the second within the village complex. An indoor man-baited net was set up in a hut near each of the outdoor nets. A sheep-baited net was also used within the village complex. This arrangement was intended to compare host preferences of each sibling species.

Preservation and identification

All mosquito specimens caught were identified visually, and members of the *An. gambiae* complex were labelled and stored individually in liquid nitrogen pending electrophoretic identification of the sibling species (MAHON *et al.*, 1976). Adults of the *An. funestus* complex were also recorded, and the adults allowed to lay eggs for larval and

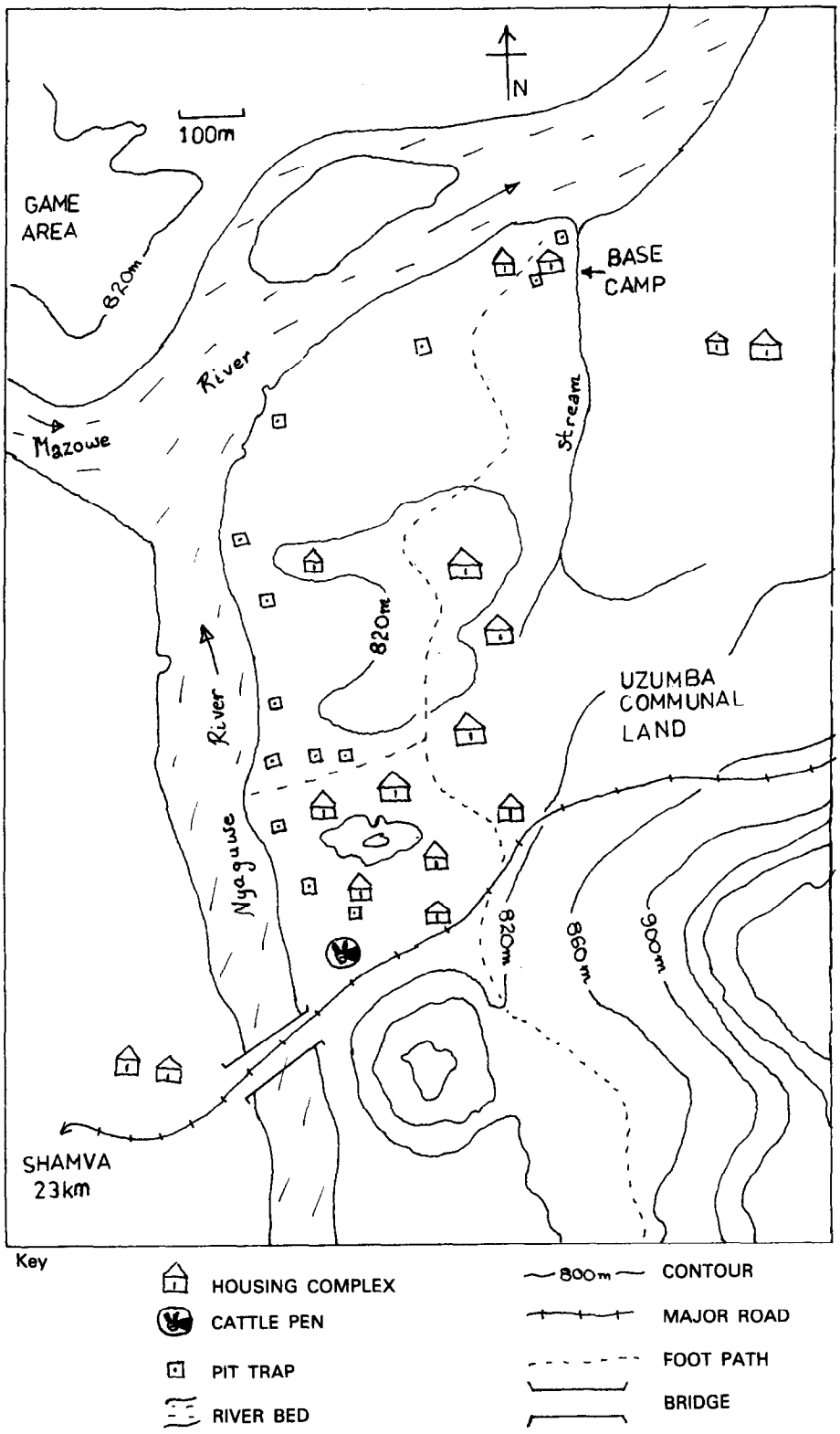


Fig. 1. Sketch map of Chitengu village.

pupal skins for separate studies of that species complex. Other species of mosquitoes were not considered to be relevant to the aims of the study, but were mounted and identified for zoogeographical distribution records.

Malaria incidence

Data for malaria notifications were obtained from rural hospitals and clinics at the nearby towns of Bindura, Shamva, Murewa, Ceres and Masembura. These centres serve the community in the neighbourhood including Chitengu village. It is to be noted that some malaria cases do not present at these centres and so actual incidence figures are likely to be higher than those reported here.

Results

Entomological findings

Although rainfall was relatively abundant (see Fig. 2a) not many *An. gambiae* sensu lato mosquitoes were captured. The peak numbers were found during March and April (Table I). Over 80% of the mosquitoes caught in pit traps and man-baited nets were accurately identified. Some of the unidentified mosquitoes had inadequate esterase enzyme activity for identification, perhaps because they were too young, while a very small proportion yielded gel patterns not consistent with those of the *An. gambiae* complex and were probably visually misidentified specimens of *An. marshalli* or other species. A small proportion of *An. gambiae* (sensu stricto) were identified but initially it was not easy to separate these from *An. quadriannulatus* Theobald on octanol dehydrogenase (ODH) enzyme patterns (MAHON *et al.*, 1976).

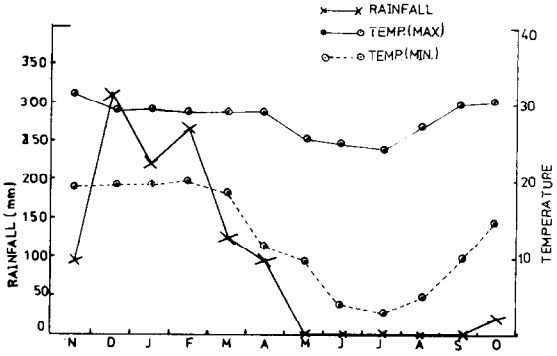


Fig. 2a. Monthly rainfall for Shamva station (Nov. 1980— October 1981) and mean monthly temperatures (°C)

Very few *An. gambiae* Giles *(sensu stricto)* were caught from the study area. The few specimens that were captured only appeared during the months of February to April and tended to be exophagic and anthropophilic since they were captured from man-baited outdoor nets (Table I). *An. arabiensis* Patton appeared around the same time as *An. gambiae* s.s. exhibiting highly exophagic and anthropophilic behaviour (see Table II). Man-baited outdoor nets yielded the highest capture for this species and could be an efficient method in sampling for *An. arabiensis*. *An. quadriannulatus* Theobald was equally as common as *An. arabiensis* in man-baited outdoor nets (Table I and II) but predominated in pit traps throughout the year.

Man-baited nets placed indoors were not useful in sampling indoor-biting mosquitoes of either species (Table II). Sheep-baited nets proved to be best for sampling *An. quadriannulatus*. The results also indicate that pit traps appear to be biased in favour of *An. quadriannulatus* whereas man-baited outdoor nets were well suited for sampling all three species. The sheep-baited net, having proved almost exclusively attractive to *An. quadriannulatus*, may need evaluation alongside calf-baited or goat-baited nets.

The captures for *Anopheles funestus* s.l., which is known to be a species complex, are summarized in Table II. During March and April both pit traps and man-baited outdoor nets produced peak numbers of mosquitoes. From May to October there was a peak population which was predominantly exophilic (pit traps) and showed no inclination to bite man in the man-baited net. However, a very high proportion of the September/October population was captured fully blood-fed.

It must be mentioned that the entomological work on *A. gambiae* s.l. was complicated by the requirements of identification by means of enzyme electrophoresis. This tends to preclude other manipulations, e.g. age-grading (DETINOVA & GILLIES, 1964), on the same specimen. The field technique of temporary storage of specimens in liquid nitrogen, as happened in this study, made them unsuitable for age-grading dissections.

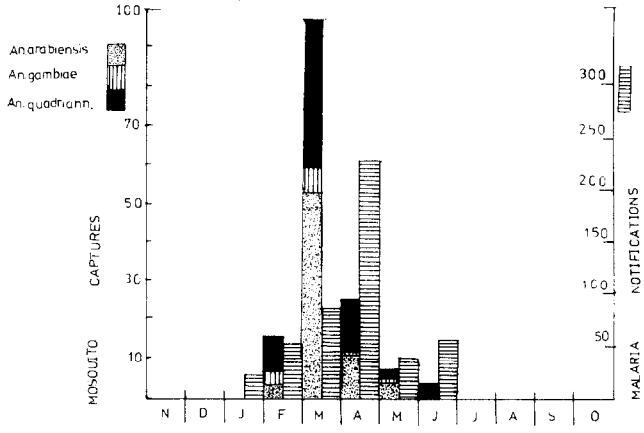


Fig. 2b. Comparison of Malaria notifications from rural hospital and clinics around Shamva with captures of the *Anopheles gambiae* complex from two human-baited trap-nets at Chitengu.

Parasitological findings

In Zimbabwe, malaria is a notifiable disease, and our laboratory receives slide submissions for confirmation of clinical diagnosis. Not all centres submit slides and slides are not submitted for all clinical malaria cases, thus malaria is probably grossly underestimated in some areas. The locally reported malaria cases for the period of this study are shown in a histogram (Fig. 2b), showing the different monthly incidences for those cases presenting at the surrounding hospitals and clinics. The histogram also compares the disease prevalence with population density of the presumed main vector *An. arabiensis*. It is to be noted that, although the absolute numbers of malaria patients are not reliable, they do show a seasonal pattern.

Discussion and Conclusions

The small numbers of *An. gambiae* s.s. coincided with the malaria transmission season and this species

Table I — Summary of *An. gambiae* complex female captures by four trapping methods (Nov. 1980 — Oct. 1981)

	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Totals	% Totals
Pit-traps														
An. gambiae	—	—	—	—	2	2	—	—	—	—	—	—	4	0,8
An. arabiensis	—	—	—	4	11	3	—	—	—	—	—	—	18	3,7
An. quadriannulatus	5	8	—	44	149	86	34	41	13	6	16	2	404	82,3
Unknown	1	12	3	4	2	32	—	4	3	2	2	—	65	13,2
Human-baited outdoor nets														
An. gambiae	—	—	—	3	6	1	1	—	—	—	—	—	11	6,1
An. arabiensis	—	—	—	5	53	11	3	—	—	—	—	—	72	40,2
An. quadriannulatus	—	—	—	8	39	13	1	3	—	—	—	—	64	35,8
Unknown	—	—	1	6	4	17	4	—	—	—	—	—	32	17,9
Man-baited indoor nets														
An. gambiae	—	—	—	—	1	—	—	—	—	—	—	—	1	
An. arabiensis	—	—	—	—	1	—	—	—	—	—	—	—	1	
An. quadriannulatus	—	—	—	—	1	—	—	—	—	—	—	—	1	
Sheep-baited outdoor net														
An. quadriannulatus	—	—	—	—	4	36	5	12	2	—	1	1	60	71
Unknown	—	—	—	1	—	11	9	—	2	1	—	—	24	29
Total Captures														
An. gambiae	—	—	—	3	9	3	1	—	—	—	—	—	16	2.52
An. arabiensis	—	—	—	9	65	14	3	—	—	—	—	—	91	14.28
An. quadriannulatus	5	8	—	52	193	135	40	56	15	6	17	3	530	83.20

Table II — Relative efficiency of four methods of sampling three sibling species of the *Anopheles gambiae* complex

		Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Totals
		n=5	n=8		n=48	n=162	n=91	n=34	n=41	n=13	n=6	n=16	n=2	n=426
Pit traps	*An. arabiensis*	—	—	—	8·3	6·8	3·3	—	—	—	—	—	—	4·2
	An. gambiae	—	—	—	—	1·2	2·2	—	—	—	—	—	—	0·9
	An. quadriannulatus	100	100	—	91·7	92·0	94·5	100	100	100	100	100	100	94·8
Man-					n=16	n=98	n=25	n=5	n=3					n=147
baited	*An. arabiensis*	—	—	—	31·3	54·1	44·0	60·0	—	—	—	—	—	49·0
nets	*An. gambiae*	—	—	—	18·7	6·1	4·0	20·0	—	—	—	—	—	7·5
(outdoor)	*An. quadriannulatus*	—	—	—	50·0	39·8	52·0	90·0	100	—	—	—	—	43·5
Man-						n=3								n=3
baited	*An. arabiensis*	—	—	—	—	33·3	—	—	—	—	—	—	—	33·3
nets	*An. gambiae*	—	—	—	—	33·3	—	—	—	—	—	—	—	33·3
(indoor)	*An. quadriannulatus*	—	—	—	—	33·3	—	—	—	—	—	—	—	33·3
Sheep-						n=4	n=36	n=5	n=12	n=2		n=1	n=1	n=61
baited	*An. arabiensis*	—	—	—	—	—	—	—	—	—	—	—	—	—
nets	*An. gambiae*	—	—	—	—	—	—	—	—	—	—	—	—	—
(outdoor)	*An. quadriannulatus*	—	—	—	—	100	100	100	100	100	—	100	100	100

Table III — Summary of *An. funestus* group captures by different trapping methods (Nov. 1980 — Oct. 1981)

Method of Capture		Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Total
Pit traps	(a)	35	2	5	29	84	72	20	12	14	27	145	73	518
	(b)	9	1	0	3	14	19	3	4	4	4	8	2	71
Total		44	3	5	32	98	91	23	16	18	31	153	75	589*
Man-baited outdoor nets	(a)	3	1	2	1	62	19	9	1	—	—	4	—	102
	(b)	—	—	—	—	—	—	—	—	—	—	—	—	—
Total		3	1	2	1	62	19	9	1	—	—	—	—	102*
Indoor Man--baited nets	(a)	—	—	—	—	—	—	—	—	—	—	—	—	—
	(b)	—	—	—	—	—	—	—	—	—	—	—	—	—
Sheep-baited net	(a)	—	—	—	2	10	23	7	5	7	2	12	12	80
	(b)	—	—	—	—	—	1	—	—	—	—	3	—	4
Total		—	—	—	2	10	24	7	5	7	2	15	12	84*
Totals		47	4	7	35	170	134	39	22	25	33	172	87	775

(a) = Fed
(b) Unfed
* = Totals for each trapping method

must be suspected as an important vector in view of its high vectorial capacity in all parts of Africa (WHITE, 1974). Its absence in most parts of Zimbabwe can be explained on the basis of its endophilic tendency (SERVICE, 1970; WHITE *et al.*, 1972) and hence its suppression by indoor DDT spraying for malaria control. The study area was last sprayed in 1972 with gamma-BHC. Subsequently, other areas in the neighbourhood were sprayed with DDT but the study area was excluded.

An. quadriannulatus was abundant in pit traps and occurred with *An. arabiensis* and *An. gambiae* s.s. in the man-baited nets both indoors and outdoors. From Table II it would appear that bait nets, especially when sited outdoors, are a more sensitive sampling method for assessing *An. arabiensis* populations than a series of pit traps. Pit traps and bait nets appear to be highly productive in sampling wild populations of *An. quadriannulatus*.

An. arabiensis and *An. gambiae* populations rose and fell in a dramatic manner, with more being captured in March than the rest of the year. On the other hand, *An. quadriannulatus* populations follow a more gradual fluctuation pattern and were present all year round. One presumes that the latter species is more tolerant of adverse weather conditions such as those that prevailed at the study site over the winter months, May to August (Fig. 2a). Some *An. quadriannulatus* specimens occur in Harare (altitude 1539 metres) during this period of the year, although the area is regarded as free from malaria and temperatures occasionally fall below 0°C during winter. *An. quadriannulatus* also occurs at higher altitudes in Ethiopia but is not a vector there (WHITE *et al.*, 1980).

At the outset of the study, just before the first rains, attempts were made to locate the breeding sites of *An. gambiae* s.l. mosquitoes. The small puddles of riverside water left as a result of gold searching were teaming with anopheline larvae, some of which were reared to adults of *An. quadriannulatus* and some *An. pretoriensis* Theobald. When *An. gambiae* and *arabiensis* appeared in February the larvae were found cohabiting the same breeding sites with *An. quadriannulatus*. This observation leads one to infer that *An. gambiae* and *An. arabiensis* probably move gradually from low altitude holoendemic malaria areas at the onset of heavy rains in February, as observed by LEESON (1931) for *An. gambiae* sensu lato. This could explain their late appearance in higher altitude areas. These findings are not conclusive, however, and the dry season ecology of these three sibling species needs further assessment. Earlier studies (LEESON, 1931) showed a decline in the population of *An. gambiae* s.l. mosquitoes either as larvae or adults in the same locality in October to early January. Therefore the present observations of seasonal abundance of *An. gambiae* s.l. can be said to be still consistent with previous findings.

The specific resting behaviour and host preferences have been thoroughtly reviewed by CHAUVET *et al.* (1969) and WHITE (1974), who pointed out that *An. quadriannulatus* is relatively harmless, being zoophilic and exophilic at all times of the year. Most *An. quadriannulatus* captures were from pit traps (Tables I and II). This indicates an exophilic tendency similar to that observed for this species in Ethiopia (WHITE *et al.*, 1980). On the contrary, very few *An. gambiae* or *An. arabiensis* were captured from pit traps. In comparison with *An. quadriannulatus*, *An. gambiae* and *An. arabiensis* were less exophilic. *An. arabiensis* has been described (WHITE, 1974) as having a high degree of ecophenotype plasticity and it would appear that the population in this study was most likely to be post-prandially endophilic. This is supported by the large numbers caught feeding outdoors in man-baited nets, an important criterion in deciding preferred hosts. The capture of blood-fed *An. quadriannulatus* from a pit trap adjacent to the cattle pen (Fig. 1) is consistent with the zoophilic status of this species and the exclusive capture of this species from the sheep-baited net is further evidence (Table II).

An. funestus exhibited two population peaks during the study period. The September/October *An. funestus* were almost entirely captured from pit traps whereas in March/April there were roughly equal captures from man-baited outdoor nets and pit traps. This would imply that at least two sibling species of the *An. funestus* complex, with contrasting host preferences, may have been present in the area. As the man-biting *An. funestus* reached a peak around the same time as the peak of malaria prevalence (Fig. 2b) the possibility exists that *An. funestus* s.s. may be playing a significant role in malaria transmission in the area, as it always does elsewhere in Africa (GILLES AND DE MEILLON, 1968).

One of the aims of this work was to correlate the seasonal incidence patterns of malaria in Zimbabwe (TAYLOR & MUTAMBU, 1981) with the vector population densities. From the graph (Fig. 2b) and Tables I to III it can be seen that malaria incidence is related to the population density of *An. arabiensis*, *An. gambiae* and one peak of *An. funestus*. The peak of malaria in April is consistent with the population peaks of these three species in March and with the fact that by April this population of mosquitoes becomes old enough for the process of sporogony to have been completed in many individual vectors. Over-all conditions obtaining during March/April could be regarded as suitable for the survival of all the different mosquito species so that malaria transmission would be at its best, whether by a single vector species or jointly with others. Elsewhere, such as in Tanzania and Kenya, it has been found that the sporozoite rate is much higher in *An. gambiae* than in *An. arabiensis* (WHITE *et al.*, 1972; HIGHTON *et al.*, 1979).

However, HARWIN (1974), in his data of malaria blood slide surveys in this country for the year 1972, observed some resurgence of malaria in October compared to the preceding months of July to September when parasite rates fell below 1%. This could be due to a number of reasons, such as sub-clinical cases of *P. malariae* recrudescing, or transmission by the *An. funestus* group or even a vectorial contribution by *An. quadriannulatus*.

From the foregoing discussion, the following observations on the malaria situation can be made. Malaria is markedly seasonal in the study area and closely related to the population densities of *An. gambiae*, *An. arabiensis* and perhaps *An. funestus*. Conditions are unsuitable for the survival of these three species at the study site for most of the year, although *An. quadriannulatus* is perennial and zoophilic, so *An. gambiae* s.s. and *An. arabiensis* probably move up from the lower, warmer zones with the onset

of the rainy seasons after the winter. Further studies are required to implicate the sporozoite-positive species and to assess their relative vectorial capacity. The precarious situation of vectors in this area causes the seasonal malaria incidence to be greatly affected by minor climatic changes.

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